

Learning the Medial Axis

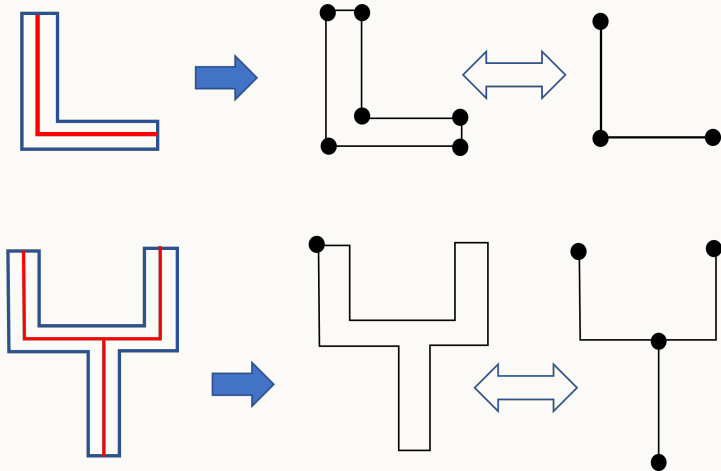
Why a neural network can find
the middle of a shape —
but not how it connects.

Yogesh H. Kulkarni · github.com/yogeshhk/MidcurveNN

The problem

ONE SENTENCE

Given a closed 2D shape, find the polyline running down its middle.



Sounds narrow. It is not.

Why it matters

Every thin part headed for structural simulation gets reduced to a simpler idealization first.

Done well: **10x cheaper** simulation.

Done badly: **silently wrong** physics.

In production CAD this is still
largely done by hand.

Attempt 1 — pixels

RASTERIZE, LEARN IMAGE TO IMAGE

It works. Seven encoder–decoder variants, UNet strongest.

But the output is pixels.

Getting a clean polyline back out of a fuzzy raster is a second unsolved problem.

Exact coordinates, connectivity, branch points — all discarded at the input.

Attempt 2 — text

SERIALIZE TO JSON, FINE-TUNE AN LLM

Best performer so far. Exact coordinates, explicit topology.

But it learns a serialization convention, not a geometric operation.

No invariance guarantees under rotation or scale.

And it takes **7 billion parameters** for a problem whose intrinsic dimension is tiny.

Both route a

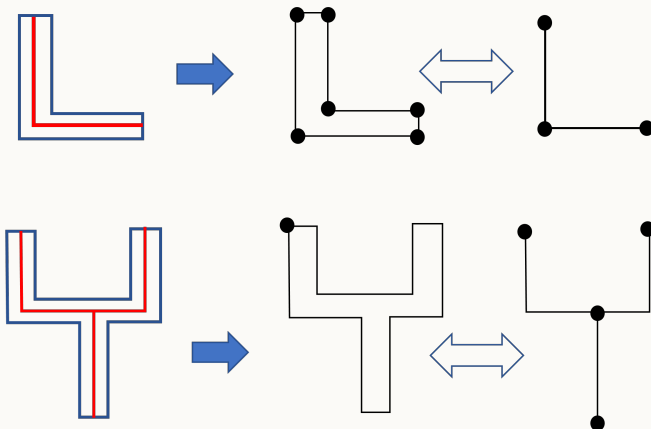
geometric problem

through a representation
that is not geometric.

Attempt 3 — geometry

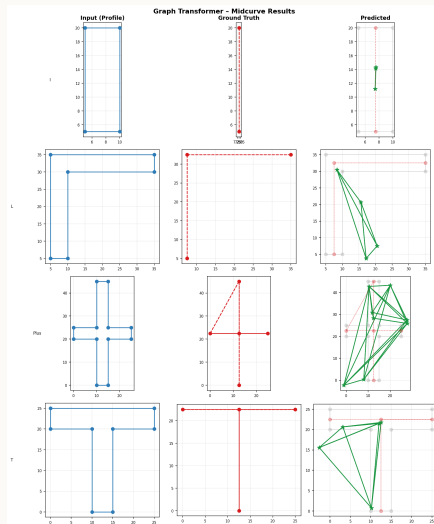
PREDICT GEOMETRY AS GEOMETRY

Shape in as a **graph**. Midcurve out as a **graph**.
Exact coordinates throughout.



Branching is native. Invariances can be built in.

What happened



It learns **where** the points go.

It does **not** learn how they connect.

Adjacency cross-entropy ≈ 1.15 — indistinguishable from chance.

Not a tuning problem

OPEN QUESTION 1

Output sets of unknown size, with no correspondence.

Nothing tells the loss which predicted node matches which true one. Chamfer distance sidesteps it and gives no gradient for topology.

OPEN QUESTION 2

Topology as a real target.

A midcurve is connected, essentially acyclic, and its branch points carry engineering meaning. Independent Bernoulli entries encode none of that.

Not a tuning problem

OPEN QUESTION 3

Long-range geometric reasoning.

A midcurve point is equidistant from two *opposing* walls — maximally far apart in graph distance. Three rounds of message passing cannot pair them.

OPEN QUESTION 4

Generalization from four shapes.

The dataset has four base topologies. Every reported number measures interpolation, not generalization. Honest evaluation needs leave-one-shape-out and synthetic data.

Two ways in

MASTER'S — ONE YEAR

Questions **1** and **4**.

Set-prediction decoder with Hungarian matching; synthetic data generator; the first honest generalization number for this problem.

DOCTORAL — THREE TO FOUR YEARS

All four, plus **midsurfaces**.

Closed to open, manifold to non-manifold. A thin solid reduces not to one sheet but to several meeting at non-manifold junctions. Learning maps whose output topology differs structurally from their input.

LOOKING FOR STUDENTS

What you get

Three implemented approaches
to benchmark against.

A working evaluation harness.

A blunt written record of everything
currently broken — down to file and line.

github.com/yogeshhk/MidcurveNN

Two-page topic summary in the repo. Students and co-supervisors welcome.